Adaptive Optics Structural Analysis for the LMT (Large Montsec Telescope)

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Disclaimer

In this deliverable AI has been used to help in code structure (e.g. to make the script modular) and in the development of the report (e.g. not to write the report but to help formatting the latex code or to correct grammar errors).

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1 Introduction

1.1 Aim of the report

This project is focused on an Adaptive Optics system, hypothetically installed in an astronomical telescope. The objective of this system would be to actively counteract the deforming effects of the atmosphere on the light.

1.2 Description of the structure and boundary conditions

The structure consists of a telescope lens of radius $R = 75$ mm. The lens is mounted on three different supports, and it is attached to several lineal actuators.

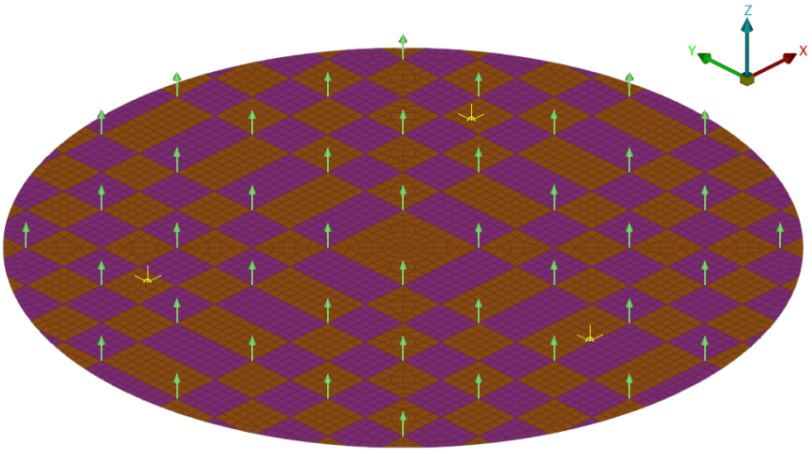


FIGURE 1: FEM model of the mirror. From: [1].

1.3 Methodology

A modular MATLAB script has been developed in this project. Each problem is implemented as an independent script, and a main driver executes them in sequence. All modules load only the data they require, and intermediate results are stored in `.mat` files to ensure clean data flow and reproducibility. META [2] has been used to visualize certain parameters.

`problem1_gravity` computes the gravity sag. Dirichlet and Neumann DOFs are identified using `findBCIndicies`, and the global gravity load is assembled as $F = Mg$. The constrained static equilibrium is solved with `solveStatics`, producing nodal displacements and support reactions. The deformed mirror shape is exported to META for visualization.

`problem2_influence` computes the influence matrix. Zernike polynomials are evaluated at normalized coordinates $(x/r, y/r)$ to form the matrix Z. Each actuator produces a 1 N load in the positive z-direction, and the resultant displacement fields are stored columnwise in U. The influence matrix is obtained from the least-squares method $C = Z \backslash U$.

`problem3_optimization` performs the force optimization. A target Zernike vector is built for modes 4–15, excluding rigid-body modes. The reduced system $C_{\text{opt}} f = Z_{t,\text{opt}}$ is solved by the least-squares method to obtain the actuator forces. Residuals are evaluated both in Zernike space and on the nodal surface after removing rigid-body components.

`problem4_modes` computes the modal reaction of the mirror. Unconstrained and constrained eigenproblems are solved using `eigs`. The first frequencies are reported, and the 6th constrained mode is exported to META.

`problem5_dynamic` evaluates the dynamic compliance and usable bandwidth of actuator 13. Modal masses, stiffnesses, and damping coefficients are computed from the constrained modes. A unit force is applied at the actuator DOF, and the modal dynamic response is evaluated over 0–2000 Hz (the double of the asked ranged). The gain curve $A(f)$ is computed, and the +3 dB crossing is used to estimate the actuator's usable bandwidth.

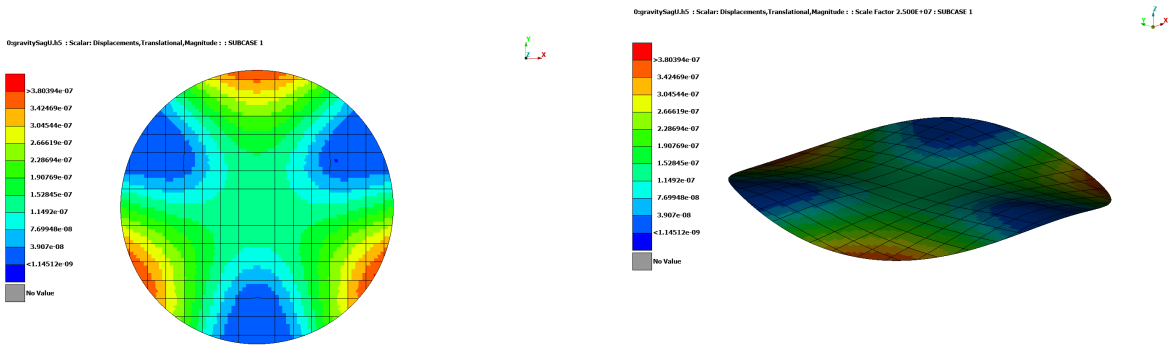
2 Gravity sag

Compute and report the gravity sag in nm-rms of the DM simply supported on its three fixations with gravity ($g = 9.81\,\mathrm{m\,s^{-2}}$) normal to the surface. You can assume the mesh spacing is uniform. Report the deformed surface map using META.

2.1 Reaction forces at each support

Reaction forces are computed for this section, yielding the following result:

Support 1 reaction	1.362900 N
Support 2 reaction	1.362639 N
Support 3 reaction	1.362989 N
Sum of reactions	4.088528 N



(A) Gravity sag: colour map only.

(B) Gravity sag: deformed configuration with colour map.

FIGURE 2: Gravity sag of the DM simply supported on three fixations visualised in META [2]. (a) colour map only; (b) deformed configuration with colours.

2.2 Mass of the model and comparison to the reaction forces

The total mass of the structure has been computed, yielding:

$$m = 4.1611 \cdot 10^{-4}\ \mathrm{t}$$

The total gravitational load computed from the mass is $W = 4.082085$ N, so the expected reaction per support (ideal equal sharing) is

$$W/3 = 1.360695 \text{ N}$$

Compared to the sum of the reactions, this result is consistent.

3 Influence matrix of the actuators and Zernike modes

Compute the influence matrix of the DM, for each actuators force and for Zernike modes.

3.1 Zernike polynomials over the surface

Zernike polynomials (Noll indices 1–100) were evaluated at each node using normalized coordinates $(x/r, y/r)$ and assembled columnwise into the matrix Z.

```
n.noll = 100;
Z = zeros(n.nodes, n.noll);
for j=1:n.noll
    [~, ~, ~, Z(:, j), ~] = zernike_noll(node_coords(:,1)/radi,...
    node_coords(:,2)/radi, j);
end
```

3.2 Compute the mirror displacements for each actuator for a unit force

The code to compute the mirror displacements for a unit force is shown below:

```
% Calcul dels coeficients d'influència
n.act = length(case_control_sets.subcase_0_SET_1);
U = zeros(n.nodes,n.act);
C = zeros(n.noll,n.act);

for i = 1:n.act
    FT = zeros(n.dof,1);
    dof = node2DOF(case_control_sets.subcase_0_SET_1(i),n,1:3);
    FT(dof(3)) = 1;

    [Uaux,~,~] = solveStatics(inD,inN,FT,KAAX,n);
    % Mirror displacements
```

```
    U(:,i) = Uaux(3:6:end);

    C(:,i) = (Z \ U(:,i)) * 1000; % um/N
end
```

3.3 Influence matrix

Results for actuator 13 (modes 4–10):

Mode	Coefficient (μm/N)
4	-4.381055×10^{-2}
5	4.795891×10^{-2}
6	2.040952×10^{-2}
7	-3.585676×10^{-3}
8	1.072863×10^{-2}
9	-1.649809×10^{-2}
10	-3.758022×10^{-3}

It can be observed that all listed coefficients are of similarly small magnitude except for modes 7 and 10, whose coefficients are an order of magnitude smaller than the rest, indicating that the primary coma (vertical) and trefoil (oblique) Zernike modes have less impact on the deformations.

4 Force vector and residuals

4.1 Compute a force vector f that minimizes $\mathbf{Z}_c(\mathbf{f}) = C\mathbf{f} - \mathbf{Z}_t$ using the least-squares method

The force vector is computed using the following script:

```
% a) Calcular el force vector amb el mètode dels minims quadrats
clear;
load common_data.mat
load influence_matrix.mat   % C, Z, U, n

% Definim Zt tenint en compte els valors de la taula (en nm RMS) i que no
% optimitzarem els noll index < 4
Zt = zeros(n.noll,1);

Zt(4)  = 1300; Zt(5)  = -650; Zt(6)  = 650;
Zt(7)  = -350; Zt(8)  = 350;  Zt(9)  = -180;
Zt(10) = 180;  Zt(11) = -120; Zt(12) = 70;
Zt(13) = -70;  Zt(14) = -45;  Zt(15) = 45;

% Definim només amb noll index > 4
C_opt  = C(4:100, :).*1000; % nm/N rms
Zt_opt = Zt(4:100);         % nm rms

% Resolució per mètode de mínims quadrats
f = C_opt \ Zt_opt; % [n.act x 1]

% Calculem els coeficients de zernike després de compensació
Zc = C_opt*f - Zt_opt;
```

4.2 The force vector for actuators 13 to 17 in N

The computed forces for actuators 13 to 17 are:

Actuator	Force (N)
13	−3.513438
14	1.822394
15	3.181910
16	1.420836
17	−2.261687

Actuator 13 and 15 carry the largest negative and positive contributions respectively.

4.3 Total compensation residual RMS rejecting rigid body modes

This quantity is the RMS of the nodal surface displacement after applying the computed actuator forces, with the rigid-body components (Zernike Noll indices 1–3) removed.

$$\text{Total error rms (excluding rigid body modes)} = 117033.334519 \text{ nm}$$

After removing rigid-body motion, the surface still exhibits a very large RMS deviation. This may be due to high-frequency surface error or computing mistakes due to units.

4.4 Compensation residual captured by the first 100 modes excluding rigid body modes

The RMS residual captured by the first 100 Noll modes (rigid body modes excluded) is presented here:

$$\text{Residual capured by Zernike modes 4..100:} = 57.661168 \text{ nm}$$

As can be observed, the residual captured by the Zernike modes is minuscule compared to the total error, which indicates that the total error is largely composed of high-frequency residuals.

4.5 High frequency residual (Noll > 100)

The residual contribution from high-frequency modes (Noll indices > 100) is summarized as follows:

$$\text{High frequency residual} = 117033.320315 \text{ nm}$$

The total RMS after compensation is extremely large ($\approx 1,17 \cdot^{5}$ nm) while the part captured by Zernike modes 4–100 is only ($\approx 57,7$ nm). This means almost all the residual is outside the first 100 Zernike modes. This result is not ideal, and there could be problems with the units when computing the total error.

5 Modal response

Compute the eigenfrequencies and eigenmodes of the model. For each case, report the frequencies in Hz.

5.1 Unconstrained. Check that 6 rigid body modes appear in the results.

The unconstrained eigenproblem yields the following lowest frequencies (Hz):

$$
\begin{aligned}
f_1 &= 0.000 \\
f_2 &= 0.000 \\
f_3 &= 0.000 \\
f_4 &= 70.504 \\
f_5 &= 74.377 \\
f_6 &= 74.808 \\
f_7 &= 707.744 \\
f_8 &= 708.2
\end{aligned}
$$

The first six modes form a cluster of very low frequencies (0–75 Hz) while mode 7 jumps by roughly an order of magnitude. This large jump confirms that the first six modes correspond to the six rigid-body motions.

5.2 Constraining the mirror simply supported on its three fixations. Report the 6th mode shape using META.

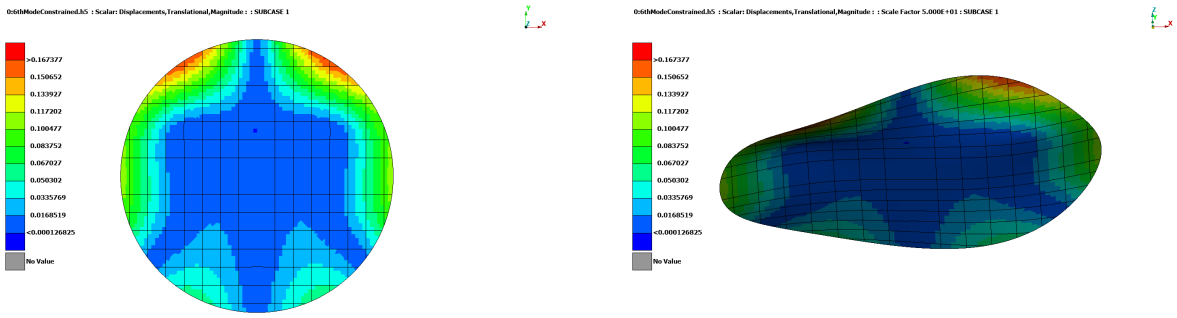
In the constrained case (Dirichlet/restricted DOFs removed) KAAX(inN,inN) and MAAX(inN,inN) are used to compute the eigenvalues and eigenvectors of the mirror.

The constrained eigenproblem yields the following lowest frequencies (Hz):

$$f_1 = 667.904 \text{ Hz}$$
$$f_2 = 668.716 \text{ Hz}$$
$$f_3 = 1074.705 \text{ Hz}$$
$$f_4 = 1454.771 \text{ Hz}$$
$$f_5 = 1782.363 \text{ Hz}$$
$$f_6 = 2037.598 \text{ Hz}$$
$$f_7 = 2039.951 \text{ Hz}$$
$$f_8 = 2941.791 \text{ Hz}$$

Constraining the three support nodes (Dirichlet DOFs removed) eliminates the near-zero rigid-body modes and shifts the spectrum upward. There can also be observed near degeneracies at $f_1 = f_2$ and $f_6 = f_7$.

The 6th constrained mode with META is shown below:



(A) Deformed configuration (colours only)

(B) Deformed configuration with mesh and colour map

FIGURE 3: Sixth constrained mode visualised in META [2]. (a) colour map only; (b) deformed configuration with colours.

6 Dynamic compliance of actuator 13 and usable bandwidth

For this part, it has been considered that the constrained modes are used. Then, after computing the modal parameters for ecah mode, the dynamic and static compliance considering a 1 N force applied by the actuator 13 are computed. The plot for the gain (A) vs the frequency is shown below:

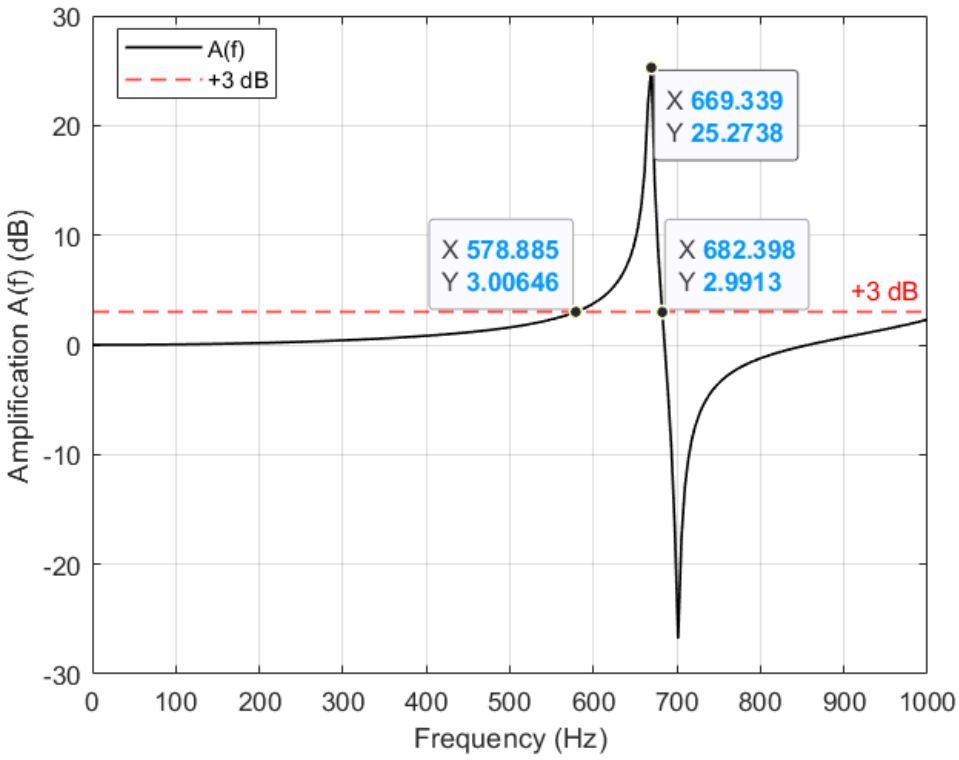


FIGURE 4: Dynamic compliance of actuator 13. Bandwidth is shown using point labels.

From Figure 4 it can be observed that in the range of 0-1000 Hz the bandwidth is computed as:

$$\text{BW} = 682.398 - 578.885 = 103.5130 \text{ Hz}$$

and that for a frequency of 669.339 there is a ressonance, yielding an amplification of 25.2738 dB, which corresponds to the first and second eigenvalues of the constrained problem.

7 Conclusions

This assignment proposed the structural analysis of a deformable mirror for an adaptive optics system. Gravity sag, actuator influence, force optimization, modal behaviour and dyanmic response were the main topics studied.

The gravity sag and reaction forces were shown to be consistent with the expected load distribution, while the influence matrix coefficients revealed that the primary coma (vertical) abd trefoil (oblique) Zernike modes were the least impactful Zernie modes regarding the deformations.

Force optimization problem yielded low error on the Zernike shape, however, the total residual error was mainly composed of high frequency components (for modes higher than 100).

Modal analysis demonstrated that the first six modes corresponded to the six rigid-body motions for the unconstrained solution for the eigenvalue problem, while the constrained solution yielded more realistic solutions.

Finally, the dynamic response for the 13th actuator identified a bandwidth usage of 104 Hz approximately, with a ressonance aligned with the first constrained modes.

Overall, the results have proven to be coherent with the expected outcome. Nevertheless, more work should be put into studying if any errors were made working with different unit systems and reference frames.

References

1. CASAMOR MARTINELL, Oriol. *Problem A statement: Adaptive Optics Structural Analysis for the LMT (Large Montsec Telescope)*. 2026. Subject: Vehicles Aeroespacials (lecture material).
2. SYSTEMS, Beta CAE. *META — Pre- and post-processing software* [https://www.beta-cae.com/meta.htm]. n.d. Accessed: 2026-03-17.